

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-a40923b7c858d708b.jsonl`  
- SHA-256: `12a9655b1dffa089f24890f7f8762cf79aadb99615635bb35b661a6aa225f51e`  
- Source modified: `2026-05-31T04:19:18+00:00`  
- Imported at: `2026-07-05T16:48:20+00:00`  
- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

Transcript

1. user (2026-05-31T04:18:22.940Z)

Source Extractor

Research question: "Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license), MinerU / PDF-Extract-Kit, RapidOCR.
- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON

transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly."

Fetch and extract key claims from this source:

****URL:**** <https://huggingface.co/allenai/olmOCR-2-7B-1025>

****Title:**** allenai/olmOCR-2-7B ? Hugging Face (Apache-2.0 weights, OCR-specialist VLM)

****Found via:**** Model-weights license minefield (commercial-safety, decisive) search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
 - be a concrete, checkable statement (not vague generalities)
 - include a direct quote from the source as support
 - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-05-31T04:18:25.328Z)

I'll fetch the olmOCR-2-7B model page and extract the relevant claims.

3. assistant (2026-05-31T04:18:28.056Z)

I have the WebFetch tool loaded. Now let me fetch the olmOCR-2-7B model page.

4. user (2026-05-31T04:18:45.737Z)

olmOCR-2-7B-1025 Model Details

1. License

****Apache-2.0****. The page states: "This model is licensed under Apache 2.0. It is intended for research and educational use in accordance with Ai2's Responsible Use Guidelines."

2. Base Model & Architecture

- ****Base Model:**** Qwen2.5-VL-7B-Instruct
- ****Size:**** 8B parameters
- ****Fine-tuned from:**** Qwen/Qwen2.5-VL-7B-Instruct
- The model card notes: "This is a release of the olmOCR model that's fine tuned from Qwen2.5-VL-7B-Instruct using the olmOCR-mix-1025 dataset. It has been additionally fine tuned

using GRPO RL training to boost its performance at math equations, tables, and other tricky OCR cases."

3. Task & Output Format

Task: Image-Text-to-Text / Document OCR and parsing

Output Format: YAML structured output (not Markdown). The example shows:

```
```yaml

primary_language: en
is_rotation_valid: True
rotation_correction: 0
is_table: False
is_diagram: False

[extracted text content]
```
```

Capabilities: Explicitly designed for tables, math equations, multi-column layouts, headers/footers, and old/damaged scans.

4. Hardware Requirements & Inference Engines

VRAM: Not explicitly stated, but model size is **8B BF16** (approximately 16GB in full precision).

Recommended Inference Engines:

- **vLLM** (primary recommendation)
- **SGLang**
- **Transformers** (via ``AutoProcessor`` and ``Qwen2_5_VLForConditionalGeneration``)
- **Docker Model Runner**

Quantization: An **FP8 version** is available (``olmOCR-2-7B-1025-FP8``), which the page recommends: "We recommend using the FP8 version for all practical purposes except further fine tuning."

Example vLLM setup:

```
```bash
pip install vllm
vllm serve "allenai/olmOCR-2-7B-1025"
```
```

5. Benchmark Scores

olmOCR-Bench Results (when used with olmOCR toolkit v0.4.0):

| Category | Score |
|---------------------|--------------------|
| ArXiv | 82.9% |
| Old Scans Math | 82.1% |
| Tables | 84.3% |
| Old Scans | 48.3% |
| Headers and Footers | 95.7% |
| Multi-column | 84.3% |
| Long tiny text | 81.4% |
| Base | 99.7% |
| Overall | 82.3 ± 1.1% |

FP8 variant scores nearly identically (82.4 ± 1.1%).

6. Release Date & Maintenance Status

- No explicit release date provided on the page
- **Version:** olmOCR-2-7B-1025 (released as of toolkit v0.4.0)

- Maintenance status: Active (associated with [olmOCR toolkit on GitHub](https://github.com/allenai/olmocr))

7. Python Integration Paths

```
**Recommended:** Use the **olmOCR toolkit**:  
```bash  
pip install olmocr>=0.4.0
```  
  
**Direct Transformers usage:**  
```python  
from transformers import AutoProcessor, Qwen2_5_VLForConditionalGeneration

model = Qwen2_5_VLForConditionalGeneration.from_pretrained(
 "allenai/olmOCR-2-7B-1025", torch_dtype=torch.bfloat16
)
processor = AutoProcessor.from_pretrained("Qwen/Qwen2.5-VL-7B-Instruct")
```
```

The toolkit provides helper functions like `render_pdf_to_base64png()` and `build_no_anchoring_v4_yaml_prompt()` for proper document preprocessing.

8. Training Dataset License

```
**SFT Dataset:** [olmOCR-mix-1025](https://huggingface.co/datasets/allenai/olmOCR-mix-1025) ?  
license not explicitly stated on this page  
**RL Dataset:** [olmOCR-synthmix-1025](https://huggingface.co/datasets/allenai/olmOCR-  
synthmix-1025) ? license not explicitly stated
```

?? ****Commercial usability:**** Not specified. The model itself is Apache-2.0 licensed but is intended "for research and educational use." Dataset commercial licensing should be verified separately at the dataset pages.

5. assistant (2026-05-31T04:18:49.508Z)

I have a thorough extraction. Let me note one important nuance for the license assessment: the model is Apache-2.0 but the card adds "intended for research and educational use in accordance with Ai2's Responsible Use Guidelines" ? this is a usage guideline overlay, not a license restriction, but it bears on the commercial-safety question and should be flagged. Let me verify the dataset license since it's directly relevant to the research question's hard constraint #3 (verify the WEIGHTS license AND training dataset license).

6. user (2026-05-31T04:18:54.098Z)

Dataset License

```
**License:** ODC-BY (Open Data Commons Attribution License)
```

```
**Commercially Usable:** Yes, with attribution required.
```

License Details

From the webpage:

> This dataset is licensed under ODC-BY. It is intended for research and educational use in accordance with Ai2's [Responsible Use Guidelines](https://allenai.org/responsible-use).

The ODC-BY license permits commercial use, but requires attribution to the dataset creators (Ai2/Allen Institute for AI).

Key Points

- ****Attribution Required:**** You must credit the dataset creators when using this data
- ****Commercial Use:**** Allowed under ODC-BY
- ****Intended Use:**** Research and educational purposes, subject to Ai2's Responsible Use Guidelines
- ****Dataset Creators:**** Ai2 (Allen Institute for AI)

For complete ODC-BY license terms, see: <https://opendatacommons.org/licenses/by/>

7. assistant (2026-05-31T04:18:58.219Z)

Both verifications are complete. This is a strong result for the research question: olmOCR-2-7B has Apache-2.0 weights AND its training dataset (olmOCR-mix-1025) is ODC-BY ? both commercially permissive, which directly clears the hard constraint #3 license minefield. The "research and educational use" language appears on both the model and dataset cards but is framed as a Responsible Use Guideline overlay, not a license restriction (the license tags themselves are Apache-2.0 and ODC-BY respectively).

This is a primary source ? the official Hugging Face model card from Allen Institute for AI (Ai2), the model's creator.

Let me record the structured output.

8. user (2026-05-31T04:19:07.294Z)

Structured output provided successfully

9. assistant (2026-05-31T04:19:18.498Z)

Done. I fetched the olmOCR-2-7B-1025 Hugging Face model card and verified the license chain.

Key findings relevant to muneem's research question:

- ****Source quality: primary**** ? official Ai2 (Allen Institute for AI) model card.
- ****License (decisive for the minefield):**** Model weights are ****Apache-2.0****. I also verified the SFT training dataset (olmOCR-mix-1025) is ****ODC-BY**** ? both commercially permissive. This is a clean pass on hard constraint #3, unlike many document-AI models (LayoutLMv3 CC-BY-NC, Surya/Marker GPL). One nuance flagged: both cards carry a "research and educational use / Responsible Use Guidelines" note, but this is a usage-guideline overlay layered on top of the permissive Apache-2.0 / ODC-BY tags, not a license restriction. I called out that distinction explicitly so the parent agent can decide whether to treat it as residual license uncertainty.
- ****Fit for bank statements:**** OCR-specialist VLM, fine-tuned from Qwen2.5-VL-7B with extra GRPO RL training targeting tables and math. olmOCR-Bench: 84.3% Tables, 84.3% Multi-column, 82.3% overall.
- ****Hardware/Python:**** ~8B BF16 (~16GB, fits the 16-24GB tier); official FP8 variant (82.4%, near-identical) shrinks VRAM for smaller GPUs. Usable via vLLM, SGLang, transformers, and the `olmocr` pip toolkit ? all local.

One caveat worth noting for the synthesis: the default output format is YAML (with `is\_table`/`is\_diagram` flags) plus extracted text ? it is a page-to-text/markup OCR model, not a native table-to-JSON cell-grid extractor, so muneem would still need to parse transaction rows out of the OCR'd text and run them through its balance-reconciliation check.

The five structured claims have been returned via StructuredOutput.